Circuit Lab Division B

Competition binder · 2026-27 season ("2027" rules) · State and National level

Daniel

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Built from the Obsidian notes in *School/Science Olympiad/Circuit Lab*. Small boxed codes after a name (like A5) tell you which tab that note is in. Page numbers are printed at the outer bottom corner of every page.

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Formulas, constants and tables, written for this binder from the checked app data and the notes.

App key card

CircuitLab key card (TI-Nspire CX II)

Home: one key opens a tool

Key	Tool		Key	Tool
o 1	Ohm's law + power		j	Kirchhoff, 2 equations
s 2	Series / parallel		y	n x n equations
d 3	Voltage divider		w	AC peak / rms / period
k	Current divider		m	Mystery resistor
l 4	LED / zener resistor		p	Battery pack (cells)
c 5	Color code (both ways)		z	USB charge time
q 6	Coulomb's law		n	Nearest E6/E12/E24
b 7	Battery capacity		h	Hit a target R from parts
e 8	Energy + cost		t	Truth table / simplify
x 9	Binary convert + add		g	Gates + De Morgan
f	Field + potential		v	Key test
a	Charge Q = It		0	Search the facts
r	Resistivity			
i	Internal resistance			
u	Fuse sizing			

Shift + letter: **N** network solver, **W** Wheatstone, **L** infinite ladder, **P** plates (E = V/d), **R** R(T), **J** energy V^2 t / R, **F** loads per fuse, **B** rated bulbs, **T** tolerance, **Q** point charges, **V** speed test.

Typing rules (all tools)

You want	Type	Example
Pick a box	its letter	`o v12 r4` -> I = 3 A
kilo / milli / micro / nano	`k m u n` after a digit (Shift+M = mega). A space before it is fine	`220k`, `20m`, `4k7`, `4.7 k`
Power of ten	`1.6*10^-19` works everywhere; `1.6x10^-19` too, except in tools with an x box (k, m, i, j, W): there use `*`; EE key; or `e` after a digit (NOT in tools with an e box: a, f, i, e, j, z — use EE there)	`1.6e-19`
Negative	`(-)` or `-` (a `-` at the end also makes it negative)	`a2u-` = -2 uC
Units after a number	allowed: `12V`, `4ohm`, `2000mAh`, `2.5Ah`, `1.2kWh`, `5min` in a time box	`bc2000mAhi.5` -> 4 h
`/` in an ohm box	parallel: `100/2` = 100 || 2 (warning). Divide: `100*1/2`	
SMD resistor code	`c472` -> "If SMD code: 4.7 kohm"; `c4R7` = 4.7 ohm	
Hours, areas, packs	`3h` in a seconds box (moves to box h); `2mm^2`, `3cm^2` in an area box; `p4S2P` = 4 series, 2 parallel. Capital M = mega (warning outside ohm boxes)	`ai2t3h` -> 21.6 kC
Math in a box	`+ - * / ^ ( )`	`v4*1.5` (4 cells), `v24-16`, `p3*60`
Series / parallel in an ohm box	`+` series, `/` parallel; with `*`, `^` or `-` in the box it is plain math (`/` divides)	`b4.7k/6.8k`, `r120^2/1000`

You want	Type	Example
Copy an answer	**enter** copies the selected box; on a box you typed it copies the first answer, enter again = the next one (the message names it)	
Paste it into another tool	`=` in a box, or ctrl+V. A digit right after `=` replaces the paste (`v=12` = 12)	`s100/100` enter, esc, `o i2 r=`
Clear / back / home	ctrl+del / del / esc	

No Enter is needed: the answer updates after every key. Empty boxes are the unknowns.

Network solver (Shift+N) shortcuts

You want	Type
Resistor 100 ohm, nodes 1-2	`1-2 100`
9 V source, + at node 1	`V1-0 9`
Commas are optional	`V1-0 9 1-2 100 2-0 200`
A chain of equal resistors	`1-2-3-4-1 5` = four 5 ohm parts around a loop
Same value as the last resistor	leave the value out: `1-5 2-6 3-7`
All resistors equal R	leave every value out: answer is in units of R
3 equal resistors in parallel	`2-0*3 6` (times key)
Before / after: part removed or shorted	`o2-3 1` (removed after) / `s2-3 1` (shorted after) is its own part: type the bulb that changes with o or s, not twice. Shows each P before -> after, x2.78 brighter
R between nodes 1 and 7	`=1-7`
Voltmeter / ammeter	`V2-3` / `I2-3` (current in the parts between 2 and 3)
Units after a value	allowed: `V1-0 9V`, `I2-0 5mA`, `1-2 100ohm`
Diode	`D2-3` = anode 2, cathode 3, 0.7 V; `D2-3 2` = 2 V LED; `D2-3 0` = ideal (ON/OFF found for you)
Zener	`Z2-3 5.1` = anode 2, cathode 3, Vz 5.1 V (forward 0.7 V); shows ON forward / ON zener / OFF
Dependent source	`V3-0 5*V1-2` (5 Vx), `I3-0 .6*I2-0` (0.6 ix, ix 2->0), `I2-0 2m*V1`; `=a-b` keeps them on
One unknown value + one condition	`2-0 x` and `I2-0=20m` (or `V2=5`, `P=2`, `=1-0 500`)

Cube, corner to corner: `N1-2-3-4-1 5-6-7-8-5 1-5 2-6 3-7 4-8 =1-7` -> 0.8333 R (5R/6).

Other text tools

Tool	Shortcut
Shift+B bulbs	`=120s` only series, `=120p` only parallel
Shift+Q charges	`e` = 1.602E-19 C (`-e` electron, `2e`); commas optional; with commas, a part with only x repeats the charge: `Qe 0, 4, -6`. 2D: `q x y` for each (commas), `=x y`: F with angle, total U, E and V
x binary	`0101^0011` XOR, `1100*1010` AND, `5 or 3`; `h9` hex, `d101` decimal; `101-11` subtract
t truth table	`^` = XOR, `-` or `~` = NOT, `+` = OR, nothing = AND; output column `0110`, with `x` = don't care: `x1x10x0100010001`; minterms `abc m1,3,5 d6`, maxterms `abc M0,2`; every table shows the mux inputs (select = first letters)

Tool	Shortcut
s series/parallel	with `*` or `^` the line is plain math (`/` divides): `s12*2^2` = 48

Search (key 0)

Type 2-5 letters of a title word: `kir`, `gfc`, `rm`. Two words: `co la` (Coulomb's law). Abbreviations work: `kvl kcl led emf rms gfci afci sop pos kmap lsb msb pot mov awg nimh lipo`. One wrong letter is forgiven (words of 4+ letters). Digits stay in the query (`e12`, `7400`); tab or an arrow moves to the list, then 1-9 opens a result. The full code list is in the binder.

Quick Reference

Densest tables first. Formulas use V = volts, I = amps, R = ohms (Ω), P = watts, t = seconds unless a box says hours. Every 2027 answer must be **metric** with **correct significant figures**. Math is algebra only: **no trig, at most 2 equations**. Capacitors, inductors, RC time constants, magnetism, transformers and Thévenin are **OUT for 2027** (a capacitor symbol can still be a wrong answer choice).

Ohm's law and power: every form

Find	From V, I	From I, R	From V, R	From P and one more
V	—	V = I·R	—	V = P/I V = √(P·R)
I	—	—	I = V/R	I = P/V I = √(P/R)
R	R = V/I	—	—	R = V²/P R = P/I²
P	P = V·I	P = I²·R	P = V²/R	—

- **Energy** E = P·t = V·I·t = I²·R·t = V²·t/R (joules if t in s).
- **Charge** Q = I·t. **Current** I = Q/t. **Voltage** V = E/Q (1 V = 1 J/C).
- **kWh** = watts × hours ÷ 1000. **Cost** = kWh × price per kWh.
- **Conductance** G = 1/R, unit siemens (S), old name mho (℧).
- Fixed R: double V → double I → **4× P**. V doubles and R halves → P × 8.
- Resistor rating: I_max = √(P_rated/R), V_max = √(P_rated·R). Common sizes 1/8, 1/4, 1/2 W.

60 W bulb at 120 V: I = 60/120 = 0.5 A, R = 120²/60 = 240 Ω. 100 W for 24 h at $0.15/kWh: 2.4 kWh → $0.36.

Series and parallel

	Series	Parallel
Shared	same **current**	same **voltage**
Adds up	voltages: $V = V_1 + V_2 + \ldots$	currents: $I = I_1 + I_2 + \ldots$
R_eq	$R_1 + R_2 + R_3 + \ldots$	$1/R_{eq} = 1/R_1 + 1/R_2 + \ldots$
Two parts	$R_1 + R_2$	$R_1R_2/(R_1 + R_2)$ (product over sum)
n equal R	n·R	R/n (series ÷ parallel = n^2)
Size	bigger than the biggest	smaller than the smallest
Conductance	$1/G = 1/G_1 + 1/G_2$	$G = G_1 + G_2$
Cells	voltages add, same mAh	mAh adds, same voltage

- **Reduce from the far end** toward the source; then go back out for each branch.
- A wire across a resistor (short) → that resistor carries 0 A: remove it. A dangling resistor carries 0 A.
- R ‖ 0 Ω = 0. R ‖ open = R. Equal parallel pair = R/2.
- **Bridge** balanced when $R_1/R_2 = R_3/R_x$: the middle resistor carries 0 A (remove it).
- **Infinite ladder** (series a first, then shunt b, repeating): R = (a + √(a² + 4ab))/2. a = 1, b = 2 → 2 Ω.

10 Ω ‖ 20 Ω ‖ 20 Ω = 5 Ω. 6 Ω ‖ 3 Ω = 18/9 = 2 Ω. 100, 200, 50, 50 Ω → 200 ‖ (100+50+50) = 100 Ω.

Dividers, internal resistance, bulbs

- **Voltage divider** (series): V_x = V_source · R_x / R_total. Biggest R gets the biggest share.
- **Current divider** (2 branches): $I_1 = I_{total} \cdot R_2/(R_1 + R_2)$. Smaller R takes the bigger current.
- **Real battery** = EMF in series with internal r. V_terminal = EMF − I·r. r = (EMF − V)/I. Short-circuit I = EMF/r.
- **Max power to the load** when R_load = r: P_max = EMF²/(4r) (efficiency then 50%).
- **Bulb brightness** = the power it actually uses. Parallel: each bulb gets full V. Series: the higher-watt bulb (lower R) is **dimmer**.
- **Open** part: I = 0 and the full source voltage appears across the gap. **Shorted** part: V = 0, R ≈ 0.
- **Mystery R** (in series with known R_k): R_x = R_k · V_x / V_k.

15 V across 1 kΩ + 4 kΩ → V(4 kΩ) = 12 V. 0.60 A into 200 Ω ‖ 300 Ω → 0.36 A in 200 Ω, 0.24 A in 300 Ω. EMF 9.0 V, 8.5 V across 10 Ω → I = 0.85 A, r = 0.59 Ω. EMF 12 V, r = 2 Ω → P_max = 18 W at R_load = 2 Ω.

Kirchhoff (KCL, KVL): step by step

KCL (current law, conservation of **charge**): current in = current out at every node. **KVL** (voltage law, conservation of **energy**): the voltage changes around any closed loop add to 0.

Sign rules for KVL: through a battery from – to + = rise (+V). Through a resistor *with* the assumed current = drop (–IR); against it = +IR. A negative answer means the current really flows the other way. Do not change the sign afterward.

Nodal method (best with 1 unknown node)

1. Pick ground (0 V), usually the bottom wire.
2. Name each unknown node voltage (V_x).
3. KCL at each unknown node: sum of (V_x – V_other)/R = 0 for every branch leaving it.
4. Solve, then find each current with I = ΔV/R.

Mesh method (2 loops)

1. Draw a clockwise current I_1, I_2 in each window.
2. KVL around each loop. A shared resistor carries (I_1 – I_2) in loop 1 and (I_2 – I_1) in loop 2.
3. Write as a·x + b·y = c, d·x + e·y = f and solve.

2 × 2 solve (Cramer): D = ae – bd, x = (ce – bf)/D, y = (af – cd)/D.

Example. 10 V source (+ up) → 2 Ω → node X. 5 V source (+ up) → 3 Ω → node X. 4 Ω from X to ground.
Nodal: (V_x–10)/2 + (V_x–5)/3 + V_x/4 = 0 → ×12: 13V_x = 80 → **V_x = 6.15 V**.
I(2 Ω) = 1.92 A into X; I(3 Ω) = 0.385 A out of X into the 5 V source; I(4 Ω) = 1.54 A. Check KCL: 1.92 – 0.385 = 1.54 ✓.
Mesh: $6I_1 – 4I_2 = 10$, $4I_1 – 7I_2 = 5$ → D = –26, I_1 = 1.92 A, I_2 = 0.385 A, shared 4 Ω: $I_1 – I_2$ = 1.54 A ✓.

Only 2 equations at Div B. Thévenin, Norton and delta-Y are OUT.

Resistor color code

Color	Digit	Multiplier	Tolerance	Temp. coeff. (ppm/K)
Black	0	×1	—	250
Brown	1	×10	±1%	100
Red	2	×100	±2%	50
Orange	3	×1 k	—	15
Yellow	4	×10 k	—	25
Green	5	×100 k	±0.5%	20
Blue	6	×1 M	±0.25%	10
Violet	7	×10 M	±0.1%	5
Gray	8	×100 M	±0.05%	1
White	9	×1 G	—	—
Gold	—	×0.1	±5%	—
Silver	—	×0.01	±10%	—
No band	—	—	±20%	—

- **4 bands:** digit, digit, multiplier, tolerance. **5 bands:** 3 digits, multiplier, tolerance. **6 bands:** 5 bands + temp. coeff.
- Tolerance band (gold/silver) goes on the **right**. Gold and silver are never digits. One black band = 0 Ω jumper.
- Memory: **B**ad **B**eer **R**ots **O**ur **Y**oung **G**uts **B**ut **V**odka **G**oes **W**ell.
- Range: R_min = R(1 – t), R_max = R(1 + t). SMD: 103 = 10 kΩ, 471 = 470 Ω, 4R7 = 4.7 Ω.

Red-violet-green-gold = 27 × 10^5 = 2.7 MΩ ±5%. Brown-black-red-gold = 1.0 kΩ ±5% (950 to 1050 Ω). Yellow-violet-black-brown-brown = 470 × 10 = 4.70 kΩ ±1%.

Standard values (E-series)

Series	Tolerance	Values per decade (× 1, 10, 100, 1 k …)
E6	±20%	1.0 1.5 2.2 3.3 4.7 6.8
E12	±10%	1.0 1.2 1.5 1.8 2.2 2.7 3.3 3.9 4.7 5.6 6.8 8.2
E24	±5%	1.0 1.1 1.2 1.3 1.5 1.6 1.8 2.0 2.2 2.4 2.7 3.0 3.3 3.6 3.9 4.3 4.7 5.1 5.6 6.2 6.8 7.5 8.2 9.1

For an LED resistor pick the next value **up** (350 Ω → 390 Ω in E12). Each E12 step is about ×1.21.

SI prefixes and units

Prefix	Sym	Factor	Prefix	Sym	Factor
tera	T	10^{12}	milli	m	10^{-3}
giga	G	10^{9}	micro	μ (u)	10^{-6}
mega	M	10^{6}	nano	n	10^{-9}
kilo	k	10^{3}	pico	p	10^{-12}
centi	c	10^{-2}	femto	f	10^{-15}

m (milli) vs M (mega) differ by 10^9. Kilo is lowercase k. 1 mm² = 10^{-6} m², 1 cm² = 10^{-4} m².

Quantity	Unit	Equals
Current I	ampere A (SI base unit)	C/s
Charge Q	coulomb C	A·s
Voltage V	volt V	J/C = W/A
Resistance R	ohm Ω	V/A
Conductance G	siemens S	A/V = 1/Ω
Power P	watt W	J/s = V·A
Energy E	joule J	W·s = V·C = N·m
Resistivity ρ	Ω·m	
Electric field E	N/C	= V/m
Frequency f	hertz Hz	1/s

A unit named after a person has a capital symbol (A, V, W) but a lowercase name (ampere, volt, watt).

Unit conversions

Unit	Value
1 kWh	3.6×10^6 J (3.6 MJ)
1 Wh	3600 J
1 mAh	3.6 C (1 Ah = 3600 C)
Wh from mAh	Wh = mAh × V ÷ 1000
1 eV	1.602×10^{-19} J
1 calorie	4.184 J (food Calorie = 1000 cal = 4184 J)
1 horsepower	745.7 W
1 hour	3600 s; 1 day = 86 400 s

mAh is charge, not energy. Convert to Wh (multiply by the battery voltage) before comparing batteries or chargers.

5000 mAh × 3.7 V = 18.5 Wh. 18 W charger at 85%: 18.5 / (18 × 0.85) = 1.2 h. 3000 mAh at 1000 mA = 3 h.

Significant figures

- Nonzero digits count. Zeros between them count (2.05 → 3). Leading zeros never count (0.0047 → 2).
- Trailing zeros after a decimal point count (2.50 → 3). In 1200 with no decimal point they are unclear: write 1.20×10^3.
- **× and ÷:** keep the fewest significant figures of the inputs. **+ and –:** keep the fewest decimal places.
- Exact counts and defined numbers (2 resistors, 1000 m/km) do not limit the answer. Round only at the end.

Coulomb's law, field, potential

Constant	Value
Coulomb constant $k = 1/(4\pi\varepsilon_0)$	8.99×10^9 N·m²/C² ($\approx 9 \times 10^9$)
Permittivity of free space ε_0	8.85×10^{-12} C²/(N·m²)
Elementary charge e	1.602×10^{-19} C (exact since 2019)
Electrons in 1 C	6.24×10^{18}
Electron mass	9.11×10^{-31} kg
Proton mass	1.67×10^{-27} kg
Planck constant h	6.626×10^{-34} J·s
Speed of light c	3.00×10^8 m/s
Air breakdown	about 3 kV/mm

- **Force** $F = k \cdot q_1 \cdot q_2 / r^2$. Like charges repel (+), opposite attract (–).
- Double r → F/4. Triple r → F/9. Double one charge → 2F. Double both → 4F.
- **Field** $E = F/q = k \cdot q / r^2$ (N/C = V/m). Points away from +, toward –. Between plates E = V/d.
- **Potential** $V = k \cdot q / r$ (scalar: add with signs). **Potential energy** $U = k \cdot q_1 \cdot q_2 / r$. **Work** $W = q \cdot \Delta V$.
- E = 0 point: between two **like** charges (closer to the smaller); outside two **opposite** charges (on the smaller one's side).
- Charge: rubbing (triboelectric) → opposite signs; contact (conduction) → **same** sign; induction → **opposite** sign. Only electrons move.
- Triboelectric order (+ end to – end): rabbit fur, glass, hair, wool, silk, paper, cotton, rubber, PVC, Teflon.

+6 μC and –7 μC, 0.50 m apart: $F = 8.99 \times 10^9 \times 6 \times 10^{-6} \times 7 \times 10^{-6} / 0.25 = 1.51$ N, attractive (1.512 N with $k = 9 \times 10^9$).

Resistivity and temperature

- **$R = \rho \cdot L / A$**; round wire $A = \pi \cdot d^2/4$. Length ×3 and area ÷2 → R × 6. Double diameter → A × 4 → R ÷ 4.
- **$R(T) = R_0(1 + \alpha \cdot \Delta T)$**. α (/°C): copper 0.0039, aluminum 0.0043, tungsten 0.0045.
- Metals: R rises with temperature. Semiconductors and NTC thermistors: R falls. Superconductor: R = 0 below a critical temperature.
- AWG: a bigger gauge number = a **thinner** wire = more R. 15 A circuit → 14 AWG, 20 A → 12 AWG.

Material	ρ (Ω·m)	Material	ρ (Ω·m)
Silver	1.59×10^{-8}	Aluminum	2.65×10^{-8}
Copper	1.68×10^{-8}	Nichrome	$\approx 1.1 \times 10^{-6}$
Gold	2.44×10^{-8}	Glass	10^{10} to 10^{14}

10 m copper wire, d = 1.0 mm: $R = 1.68 \times 10^{-8} \times 10 / (\pi \times 0.001^2/4) = 0.21$ Ω. Copper 100 Ω at 20 °C → 119.5 Ω at 70 °C.

DC and AC

- **DC**: one direction, fixed polarity. Conventional current + → –; electrons flow – → +.
- **AC**: reverses, usually a sine wave. **V_rms = V_peak/√2 ≈ 0.707·V_peak**. V_peak = 1.414·V_rms. V_pp = 2·V_peak.
- **T = 1/f**. RMS = the DC value that gives the same heating power.
- Grid uses AC because it steps up to high voltage easily: same power, less current, less I²R loss.
- At 50-60 Hz, AC is more dangerous than DC at the same voltage (fibrillation, can't let go).

Mains	V_rms	f	V_peak	V_pp	T
US outlet	120 V	60 Hz	170 V	339 V	16.7 ms
US dryer/range	240 V	60 Hz	339 V	679 V	16.7 ms
Europe	230 V	50 Hz	325 V	651 V	20 ms
Japan	100 V	50/60 Hz	141 V	283 V	20/16.7 ms

US wire colors: hot = black or red; neutral = white or gray; ground = green or bare. Outlet: narrow slot = hot, wide = neutral, round = ground. US house: two 120 V hots, hot to hot = 240 V (split phase).

DC equal to a 100 V-peak AC source: 100/√2 = 70.7 V.

Batteries and USB

Cell	Volts	Cell	Volts
Alkaline, carbon-zinc (AA, AAA, C, D)	1.5	NiMH, NiCd	1.2
Lithium coin CR2032	3	Lead-acid (per cell)	2.0
Silver oxide	1.55	Lithium-ion (full)	3.6-3.7 (4.2)
Zinc-air	1.4	9 V battery	6 × 1.5 V

Size changes capacity, not voltage. 12 V car battery = 6 lead-acid cells. Anode = oxidation = – terminal while discharging ("an ox, red cat").

USB level	Power	USB level	Power
USB 2.0: 5 V 0.5 A	2.5 W	PD: 9 V 3 A	27 W
USB 3.x: 5 V 0.9 A	4.5 W	PD: 15 V 3 A	45 W
USB-C: 5 V 3 A	15 W	PD: 20 V 3.25 A	65 W
BC 1.2: 5 V 1.5 A	7.5 W	PD: 20 V 5 A	100 W

Label "Input 100-240 V ~ 50/60 Hz" = wall side (AC). "Output" = DC levels. Power at each level = V × A.

Diodes

- Current flows when the **anode** (p side, triangle) is higher than the **cathode** (n side, bar) by about V_f.
- n-type: group 15 donors (P, As, Sb), carriers = electrons. p-type: group 13 acceptors (B, Ga, In, Al), carriers = holes. Doped material is still neutral.
- Forward bias: depletion region thinner. Reverse bias: wider; only leakage until breakdown.
- Models: ideal (short/open); **constant drop**: I = (V_source – 0.7)/R for silicon.
- **Zener**: used reverse-biased at V_Z. R = (V_in – V_Z)/I.
- Bridge rectifier: 4 diodes, loses 2 × 0.7 = 1.4 V. Half-wave: 1 diode.
- Diode test on a meter: Si 0.5-0.7 V forward, OL reverse. OL both ways = open; ≈ 0 both ways = shorted.
- Diodes and LEDs do NOT obey Ohm's law (non-linear I-V curve).

Type	V_f (forward drop)	Band gap
Silicon	0.6-0.7 V	1.1 eV
Germanium	about 0.3 V	0.67 eV
Schottky	0.2-0.4 V	(metal-semiconductor)
GaAs (infrared LED)	1.2-1.5 V	1.42 eV

LEDs

LED color	Typical V_f	LED color	Typical V_f
Infrared	1.2-1.5 V	Green (InGaN)	about 3.0 V
Red	1.8-2.1 V	Blue	2.7-3.3 V
Orange / yellow	2.0-2.2 V	White	3.0-3.5 V
Green (old type)	2.0-2.2 V	UV	3.1-4.4 V

- **Series resistor: R = (V_source – n·V_f) / I_LED** (n LEDs in series). Normal I = 10-20 mA.
- Resistor power = I²R. LED power = V_f × I.
- Long leg = anode (+). Short leg and flat rim edge = cathode (–). Reverse breakdown only about 5 V.
- Shorter wavelength → bigger band gap → higher V_f. Photon energy E (eV) ≈ 1240 / λ (nm).
- Light comes from electrons recombining with holes and releasing photons.
- Equal brightness: put identical LEDs in series (same current), or give each parallel LED its own resistor.

9 V, red LED 2.0 V, 20 mA: R = 7.0/0.020 = 350 Ω → use 390 Ω (then I = 18 mA, resistor power 0.13 W). 620 nm red: 1240/620 = 2.0 eV.

Logic gates

Gate	US symbol	Expression	Out for AB = 00 01 10 11	NAND / NOR count	Chip
NOT	triangle + bubble	A′, Ā	A=0→1, A=1→0	1 / 1	7404
AND	D shape	A·B	0 0 0 1	2 / 3	7408
OR	curved shield	A + B	0 1 1 1	3 / 2	7432
NAND	D shape + bubble	(AB)′	1 1 1 0	1 / 4	7400
NOR	shield + bubble	(A + B)′	1 0 0 0	4 / 1	7402
XOR	shield + extra back curve	A⊕B = A′B + AB′	0 1 1 0	4 / 5	7486
XNOR	XOR + bubble	(A⊕B)′ = AB + A′B′	1 0 0 1	5 / 4	74266

- A **bubble** always means NOT. IEC boxes: "&" AND, "≥1" OR, "=1" XOR, "1" NOT/buffer.
- Universal sets: {NAND}, {NOR}, {AND, NOT}, {OR, NOT}. NOT universal: {AND, OR}.
- XOR from 4 NANDs: N1 = NAND(A,B); out = NAND(NAND(A,N1), NAND(B,N1)).
- Logic levels: 1 = HIGH (5 V or 3.3 V), 0 = LOW (0 V). TTL: low < 0.8 V, high > 2.0 V.
- Switches: series = AND, parallel = OR. Diode OR and diode AND work; NOT needs more than diodes.
- 14-pin chips: pin 14 = +5 V (V_cc), pin 7 = ground; count counter-clockwise from the notch. **7402 has a different pinout** (outputs on pins 1, 4, 10, 13).

Boolean algebra

Law	OR form	AND form
Identity	A + 0 = A	A·1 = A
Null	A + 1 = 1	A·0 = 0
Idempotent	A + A = A	A·A = A
Complement	A + A′ = 1	A·A′ = 0
Double NOT	(A′)′ = A	
Commutative	A + B = B + A	AB = BA
Associative	(A+B)+C = A+ (B+C)	(AB)C = A(BC)
Distributive	A + BC = (A+B) (A+C)	A(B+C) = AB + AC
Absorption	A + AB = A	A(A+B) = A
"Disappearing opposite"	A + A′B = A + B	A(A′+B) = AB
Consensus	AB + A′C + BC = AB + A′C	
De Morgan	**(A + B)′ = A′·B′**	**(A·B)′ = A′ + B′**

- De Morgan: "break the bar, change the sign". Works for more inputs: (ABC)′ = A′ + B′ + C′.
- Bubble pushing: NAND = OR with inverted inputs. NOR = AND with inverted inputs. AND-OR → NAND-NAND; OR-AND → NOR-NOR.
- Trap: in Boolean algebra 1 + 1 = 1 (OR). In binary addition 1 + 1 = 10.

Truth tables, binary, adders, MUX

- n inputs → **2^n rows** (2 → 4, 3 → 8, 4 → 16). List rows in binary counting order.
- **Sum of products**: for each row with output 1, AND the inputs (NOT the 0s); OR the terms. Rows 01, 10 → A′B + AB′ (XOR).
- Majority of 3: Y = AB + BC + AC. "Alarm if door open AND armed, OR panic": Y = D·A + P.
- Binary place values: 1, 2, 4, 8, 16, 32, 64, 128. 1011 = 11. 1011 + 0110 = 10001 (11 + 6 = 17).
- Addition: 1+1 = 0 carry 1; 1+1+1 = 1 carry 1. A carry out of the top bit = overflow.
- **Half adder**: S = A⊕B, C = A·B. **Full adder**: S = A⊕B⊕C_in, C_out = AB + C_in(A⊕B) = AB + AC_in + BC_in.
- Full adder = 2 half adders + 1 OR. 4-bit ripple-carry adder = 4 full adders (or 3 full + 1 half).
- **MUX**: 2^n data inputs, **n select lines**, 1 output. 4-to-1 → 2 lines, 8-to-1 → 3, 16-to-1 → 4 (not √N). 2-to-1: Y = $S'D_0 + S \cdot D_1$.
- DEMUX: 1 input to one of 2^n outputs. Decoder: n inputs → 2^n outputs, one on at a time. Encoder = reverse.
- Sequential logic (flip-flops, latches, counters, clocks) is OUT for 2027.

Meters and breadboards

Measure	Connect	Power	Ideal meter R
Voltage	**Across** the part (parallel)	ON	infinite (real DMM ≈ 10 MΩ)
Current	Break the circuit, **in series**	ON	0 Ω
Resistance	Across the part, part removed or one leg lifted	**OFF**	—

- Black lead in **COM**. Red in **VΩ** for volts, ohms, diode test, continuity; red in **mA** or **10 A** for current.
- **Never** put an ammeter straight across a battery: it is a short and blows the meter fuse.
- "OL" = over range or open circuit. DC volts = V with straight line and dashes; AC = V~. Start on a high range.
- Loading: a low-resistance voltmeter across a big resistor reads **low**. Fingers on both probes add body R in parallel.
- Current without an ammeter: measure V across a known R, I = V/R. Power supplied: P = V_source × I.
- Breadboard: each row of 5 holes (a-e, f-j) is connected; the center gap separates halves; chips straddle the gap. Rails run lengthwise (red +, blue –); long boards may split rails in the middle: check with continuity.
- Troubleshoot: check power, battery, switch, fuse, loose wire → expected V at each point → measure V powered → measure R unpowered.

Switches and relays

Switch	Meaning	Terminals
SPST	single pole, single throw: on/off	2
SPDT	single pole, double throw: changeover ("3-way")	3
DPST	two SPSTs moving together	4
DPDT	two SPDTs together (reverse a motor; "4-way" when wired as a crossover)	6

- Pole = number of separate circuits. Throw = positions each pole connects to. NO = normally open, NC = normally closed.
- **3-way**: two SPDT switches joined by two travelers; commons go to supply and load. Light = A XNOR B (or XOR). Each extra location adds one 4-way (DPDT crossover) between them.
- **Relay**: a small coil current moves contacts in a separate circuit. Terminals COM, NO, NC. **Flyback diode** across the coil (cathode to +) absorbs the turn-off spike.

Safety: fuses, breakers, GFCI, shock

- **Fuse**: thin link in **series on the hot wire**; melts above its rating; replace with the **same** rating. Fast-blow vs slow-blow.
- Does it blow? I = V/R_eq or I = P/V, compare with the rating. Exactly equal to the rating: does not blow.
- **Breaker**: resettable. Thermal trip (bimetal strip, slow overloads) + magnetic trip (coil, sudden shorts).
- Fuses and breakers protect **wires from fire**, not people from shock.
- **GFCI** compares hot and neutral currents; trips on a difference (leak, maybe through a person). Class A: trips at 6 mA or more, not below 4 mA; tests use 5 mA; shock-level faults clear in about 25 ms. Needed in bathrooms, kitchens, outdoors, garages, near water. Does not trip on overload or hot-to-neutral contact. Works without a ground wire. TEST monthly.
- **AFCI** detects arcing (sparking) faults.
- Ground wire carries current only during a fault, so the breaker trips and a metal case stays near 0 V. Neutral and ground are bonded only at the main panel.

Body current (60 Hz)	Effect
about 1 mA	felt as a tingle
about 5 mA	highest "harmless" level (GFCI level)
10-20 mA	let-go limit: muscles clamp
50-100 mA	heart may fibrillate
1-4 A and more	heart stops, severe burns

Body R: about 100 kΩ dry, about 1 kΩ wet. 120 V / 1 kΩ = 120 mA (deadly); 120 V / 100 kΩ = 1.2 mA (tingle). Habits: one hand, dry hands, unplug first, keep the ground pin.

History: the seven 2027 scientists

Person	Life, country	Unit	Know for
Ampère	1775-1836, French	ampere A	1820: parallel currents attract, opposite repel (after Oersted); "electrodynamics"; "Newton of electricity"
Coulomb	1736-1806, French	coulomb C	1785: inverse-square law of electric force, measured with his torsion balance
Kirchhoff	1824-1887, German	**none**	1845 (as a student): current and voltage laws; spectroscopy with Bunsen (cesium, rubidium); "black body"
Volta	1745-1827, Italian	volt V	Voltaic pile, first battery (announced 1800); electrophorus 1775; disputed Galvani
Ohm	1789-1854, German	ohm Ω	1827 "Die galvanische Kette": V = IR; used a thermocouple source
Hertz	1857-1894, German	hertz Hz	1886-89: first made and detected radio waves (proved Maxwell); saw photoelectric effect 1887
Boole	1815-1864, **English** (taught in Cork)	**none**	Boolean algebra: "Mathematical Analysis of Logic" 1847, "Laws of Thought" 1854

Others: Franklin (named + and −, kite 1752); Galvani (frog legs, "animal electricity" 1791); Oersted (1820, current moves a compass); De Morgan (laws, 1847); Shannon (1937 thesis: Boolean algebra for relay circuits); Edison (DC) vs Tesla and Westinghouse (AC); Joule (heating I^2R); Watt (steam engine, no electrical work); siemens S = Werner von Siemens. Faraday and Tesla history is OUT for 2027.

TAB S

Search codes (appendix)

History (147) 733
Static electricity (50) 733
DC and AC (41) 733
Batteries (31) 734
Basics: I, V, R, P (14) 734
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Short search codes for the CircuitRef facts on the TI-Nspire. Type a code in search (key 0).

Search codes

Type the code in CircuitRef search (key 0) and the fact is first in the list. CircuitLab (229 core facts) uses gen/codes_core.tsv — codes there may be even shorter.

A dash (—) means no short code puts that fact first: search two words of its title instead.

History

af th After the pile: first uses
len Ampere force law: lengths
am na Ampere named galvanometer
am un Ampere unit history
dis Ampere's discoveries
as Ampere: astatic needle
post Ampere: career posts
ampe Ampere: life
sep Ampere: Sept 1820 timeline
he Ampere: words he coined
co pr Before Coulomb: priority
bl no Blue LED Nobel 2014 detail
de le Boole and De Morgan letters
trea Boole quote: treatise
bo la Boole's law x^2 = x
je Boole's OR vs Jevons
sy Boole's symbols
1847 Boole: 1847 book details
books Boole: books
tho Boole: Laws of Thought title
bo Boole: life
roy Boole: Royal Medal 1844
cav Cavendish found it first
grad Coulomb quote: graduating
posth Coulomb unit: posthumous
178 Coulomb's 1785 data
co hi Coulomb's law history
att Coulomb: attraction method
cou Coulomb: life
prov Coulomb: proved vs assumed
gl Crown of cups: wine glasses
shan De Morgan and Shannon
de De Morgan: life
du Du Fay: two kinds of charge
ed Edison's argument vs AC
swa Edison: bulb 1879 and Swan
el ho Electrophorus: how it works
end End of the war 1892
fra Franklin
fr Franklin: battery and Fr unit
ju Franklin: kite June 1752
ga Galvani
179 Galvani: 1791 book title
gi Gilbert: De Magnete 1600
verso Gilbert: electricus, versorium
gu Gray and von Guericke
qu Hertz quotes
unit Hertz unit history
187 Hertz: 1879 Berlin prize
ray Hertz: cathode rays
her Hertz: life
1887 Hertz: photoelectric 1887
rad Hertz: radio waves
rece Hertz: the receiver
he tr Hertz: the transmitter
show Hertz: wave properties shown
hol Holonyak: first visible LED
jo Joule and Watt
pad Joule: paddle wheel
whil Joule: unit named while alive
1857 Kirchhoff 1857 signal speed
sun Kirchhoff quote: gold sun
ki di Kirchhoff's discoveries
stu Kirchhoff: 1845 as a student
bla Kirchhoff: black body
ces Kirchhoff: cesium and rubidium
cit Kirchhoff: cities
ali Kirchhoff: law aliases
kir Kirchhoff: life
spect Kirchhoff: spectroscope trap
ki th Kirchhoff: the other laws
maxw Maxwell: link to Hertz
mh Mho and siemens history
new Newton of electricity quote
oe Oersted
pap Oersted: paper and Arago
pn Ohl: PN junction details
mem Ohm quote: the Memoir
oh un Ohm unit name history
182 Ohm's 1826 formula
aco Ohm's acoustic law
hi Ohm's law history
die Ohm: Die galvanische Kette
ho Ohm: honours
fou Ohm: inspired by Fourier
li Ohm: life
pog Ohm: Poggendorff thermocouple
1827 Ohm: the 1827 controversy
ley Pile vs Leyden jar
pn le PN junction and LED history
names Seven names mnemonic
det Shannon thesis details
194 Shannon: info theory 1948
195 Solar history 1839, 1954
stead Steady-state laws multi-select
tesl Tesla (out for 2027)
tha Thales and amber c. 600 BC
sci The seven 2027 scientists
timel Timeline of the seven
185 Timeline: 1850 to 2014
bc Timeline: 600 BC to 1850
ba ho Torsion balance: how it works
flu Trap: Ampere and fluorine
fo ma Trap: Ampere found magnetism
tr am Trap: Coulomb or Ampere?
el ke Trap: electrodynamics key
mix Trap: Galvani mix-ups
jo oh Trap: Joule before Ohm
stru Trap: kite struck by lightning
nob Trap: Nobel starts 1901
tr wo Trap: not Boole's work
tr ki Trap: not Kirchhoff's work
tr vo Trap: not Volta's
fara Trap: Oersted law = Faraday
bis Trap: Ohm's copper-bismuth
riv Trap: Ohm's rival not Galvani
mag Trap: On Magnetism key
pred Trap: predicted vs confirmed
sod Trap: sodium in Sun, not Moon
tor Trap: torsion balance vs Ohm
mad Trap: Volta made electrophorus
worke Trap: Volta worked in France
whi Trap: which Siemens
wh el Trap: who coined electricity
inven Trap: who invented AC
cong Units at 1881 Paris congress
aft Units named after people
188 Volt unit named 1881
honor Volta honors timeline
180 Volta letter: 1800 details
exp Volta quote: experiment
vo vs Volta vs Galvani: details
qs Volta vs Galvani: past Qs
vo co Volta's contact series
prop Volta's law: Q prop. to V
volta Volta: life
17 Volta: methane 1776
ot Volta: other work
pil Voltaic pile
vo bu Voltaic pile: build details
war War of the Currents
ado Watt and joule adopted 1889
hp Watt: steam engine, hp
wer Werner von Siemens
wes Westinghouse: AC wins
whea Wheatstone bridge history
pe Zener: person vs diode

Static electricity

kv Air breakdown 3 kV/mm
car Car safe in lightning
cons Charge constants
quan Charge quantization
meth Charging methods
cond Conductors in fields
insu Conductors vs insulators
ratio Coulomb ratio: charges
dist Coulomb ratio: distance
co un Coulomb unit = 1 A*s
co la Coulomb's law
disch Discharge types order
el ch Electric charge basics
el fi Electric field
39 Electric vs gravity 10^39
el Electron-volt
el ind Electroscope by induction
leaf Electroscope leaf rules
equi Equipotential rules
es ESD you cannot feel
cag Faraday cage: charge inside
fi Field lines
whe Field where E = 0
fla Flash-to-bang rule
dir Force direction in a field
ig Fuel ignition risk
flo Grounding electron flow
hu Humidity and static
tou Identical spheres touch
indu Induction steps in order
joi Joined spheres: Q ~ r
un k value and units
nu Lightning numbers
3rd Newton 3rd law: equal force
be Pith ball behavior
po Polarization
po en Potential and energy
qua Quark charges
sta Static hazards
cau Static shock cause
thu Thunderstorm charging
ve Tribo series: long version
mos Tribo: most charge pair
trib Triboelectric series
tw sp Two touching spheres induced
unif Uniform field E = V/d
van Van de Graaff safe to touch
wim Wimshurst machine
moh Wrist strap 1 Mohm resistor
ze po Zero-potential points: two

DC and AC

240 120 V and 240 V split phase
ac sy AC and DC symbols
ac AC basics
da AC vs DC danger
ad la Adapter label symbols
ben Bench supply CV and CC

`buc` Buck, boost, DC-DC
`co` Commutator
`ar` DC arcs harder to stop
`dc` DC basics
`ha` DC hazards
`dc us` DC sources and uses
`grov` Fuel cell: Grove 1839
`hot` Hot, neutral, ground roles
`hou` Household wire colors
`hv` HVDC lines
`60` Japan mains 100 V 50/60 Hz
`soc` Lamp socket: shell is neutral
`los` Line loss and voltage
`mai` Mains in other countries
`ne` NEMA name code
`ou me` Outlet meter readings
`ou` Outlet slots
`rm` RMS
`sola` Solar cell numbers
`ac or` Source: AC or DC?
`seeb` Thermocouple: Seebeck 1821
`34` Trap: 340 V peak-to-peak
`120x` Trap: 60 Hz reverses 120x/s
`les` Trap: AC has less resistance
`tr ad` Trap: adapter vs converter
`all` Trap: all AC is a sine wave
`10 pe` Trap: is 100 V peak or RMS?
`125` Trap: rated 125 V not 120 V
`sq` Trap: RMS of a square wave
`nev` Trap: wave never below zero
`110` US 110, 115, 117, 120 V
`us` US and world outlets
`wal` Wall adapter insides
`gri` Why the grid uses AC
`wi` Wire gauge AWG

Batteries

`al` Alkaline cell
`ano` Anode and cathode
`ra` C-rate
`ca en` Capacity and energy
`zi` Carbon-zinc (Leclanche)
`cc` CC/CV slow finish
`cells` Cells in series and parallel
`adds` Cells: r in series, parallel
`cr` CR2032 lithium coin
`18` Daniell cell 1836
`dr` Dry cell vs wet cell
`eve` Everyday source voltages
`int` Internal resistance
`lea` Lead-acid cell
`lem` Lemon or potato cell
`lip` LiPo and LiFePO4
`io` Lithium-ion cell
`max` Max power transfer
`ni` NiCd cell
`nim` NiMH cell
`3s` Pack names like 3S2P
`pa ce` Parts of a cell
`pr ce` Primary cell voltages
`prim` Primary vs secondary
`rech` Rechargeable cell voltages
`stor` Trap: battery stores charge
`used` Trap: copper is used up
`24` Trap: diesel battery 24 V
`nai` Trap: two copper nails = 0 V
`ty ba` Typical battery capacities
`wea` Weak battery reads full

Basics: I, V, R, P

`co fl` Conventional vs electron flow
`cu` Current
`dri` Drift speed of electrons
`ef` Efficiency
`cos` Electricity cost
`em` EMF is not a force
`en` Energy and kWh
`en co` Energy conversions
`la` Ohm's law
`sc` Out of scope 2027
`pow` Power formulas
`ap` Typical appliance powers
`typi` Typical currents
`vo` Voltage

Units

`19` 19 SI units named for people
`am ba` Ampere is the base unit
`am si` Ampere: only SI base unit
`non` Non-SI conversions
`pai` Prefix shortcut pairs
`qu un` Quantities and units table
`sev` Seven SI base units
`si oh` SI base units of V, ohm, S
`si` SI prefixes
`fig` Sig fig counting rules
`sig` Sig figs in math
`tr el` Trap: 1 A = 1 electron/s
`tr la` Trap: Coulomb law vs unit
`plu` Trap: S vs s, plurals
`tr wa` Trap: watt vs joule
`wr` Unit writing rules
`un eq` Units equal to a volt
`un wa` Units equal to a watt or ohm

Resistors

`zer` 3-band and zero-ohm
`sy li` Anti-symmetry: V/2 line
`cop` AWG copper table
`sca` AWG scaling rules
`bala` Balanced Wheatstone bridge
`fil` Bulb filament and inrush
`dig` Color code digits
`gol` Color code gold silver
`mn` Color code mnemonic
`del` Delta-Y conversion
`e1` E12 standard values
`e6` E6 and E24 series
`eq va` Equal-resistor values
`eq` Equivalent resistance tips
`far` Finite ladder: far end first
`ite` Infinite ladder 0.618R
`61` Infinite ladder 1.618R
`metal` Metal resistivity order
`to` Network tolerance
`ohmi` Ohmic vs non-ohmic
`tap` Pot taper: linear vs log
`pot` Potentiometer
`po rh` Potentiometer vs rheostat
`str` Power limit of a string
`re ba` Reading 4, 5, 6 bands
`resi` Resistivity
`re va` Resistivity values
`meta` Resistivity: non-metals
`fe` Resistivity: W, Fe, Pt, Pb
`cub` Resistor cube values
`re po` Resistor power rating
`re ch` Resistor tolerance check
`rh` Rheostat
`gr` Slope of V-I graph
`bun` Stretch or bundle a wire
`su di` Superconductor discovery
`cut` Symmetry: join or cut
`6t` Tempco band (6th band)
`res` Temperature and resistance
`coe` Temperature coefficient
`ld` Thermistor, varistor, LDR
`tr pa` Trap: 1/8 W parts
`unb` Unbalanced bridge voltmeter
`read` Which end to read first
`nich` Why heaters use nichrome

DC circuits

`ie` ANSI vs IEC symbols
`ba co` Battery symbol: count pairs
`pa dimm` Bulb A then B, C in parallel
`bul` Bulb brightness
`bu ra` Bulb rating to resistance
`cu di` Current divider
`dea` Dead lamp: 0 V or full V
`ea` Earth ground symbol
`fu sy` Fuse and breaker symbols
`pa` How parts fail
`rank` Identical bulbs: rank rules
`lam` Lamp symbol: X or loop
`loade` Loaded divider: output drops
`op` Open circuit
`para` Parallel resistors
`sc ba` Schematic symbols: basics
`sch` Schematic symbols: parts
`pa ra` Series / parallel ratio n^2
`ser` Series resistors
`nodes` Series vs parallel by nodes
`sh` Short circuit
`wha` Shorted bulb: what happens
`bra` Shorted parallel branch
`sh se` Shorting one series bulb
`opens` Switch closes or opens
`th sy` Thermistor and LDR symbols
`div` Voltage divider
`ze cu` Zero-current branch trick

Kirchhoff

`deli` Absorb or deliver power
`fixe` Battery to ground fixes a node
`un pi` Count unknowns, pick method
`kc` KCL current law
`ru` KVL sign rules
`kv la` KVL voltage law
`me` Mesh analysis
`nod` Nodal analysis
`mil` One-node formula (Millman)
`su` Supernode
`steps` Superposition steps
`thev` Thevenin and Norton: OUT
`tr ju` Trap: junction rule = charge

Meters

`accu` 3.5-digit meter, accuracy
`am` Ammeter
`bur` Ammeter burden
`box` Black box: find the diode
`conti` Continuity and diode test
`di te` Diode test readings
`jack` Fuse in the mA jack
`galvano` Galvanometer
`idea` Ideal meters memory trick
`ind` Indirect measurements
`ke` Kelvin bridge
`le ze` Lead resistance zero
`le fa` LED fault from voltages
`loa` Loading effect
`me po` Meter polarity: minus sign
`se` Meter settings
`lett` Meter symbols: letters
`ja` Multimeter jacks
`oh` Ohmmeter
`pic` Pick the best range
`red` Red lead is anode
`shu` Shunt vs multiplier
`am le` Trap: ammeter 0, LED lit
`lef` Trap: lead left in mA jack
`voltm` Voltmeter
`vo se` Voltmeter in series

Hands-on tasks

`wa bu` 3-way build checklist
`wa fa` 3-way fault tables
`lay` Breadboard layout
`lim` Breadboard limits
`br ro` Breadboard rows of 5
`33` BullSO C 133.33 ohm task
`unc` Carry uncertainty
`ch` Charge time

co br Coin bridge switch test
le ch Equal LEDs checklist
fa Fault clues
commo Find the SPDT common pin
poi Hands-on scoring points
shar Hands-on share of score
hug Huge mystery R: loading
mini Minimum safe resistor
my Mystery R checklist
my re Mystery resistor task
pc PCB printed circuit board
clo Pick R_k close to R_x
bank Power bank real output
supp Power supplied task
po ch Power task checklist
varie Rail polarity varies
ont Schematic onto breadboard
so Sources of error
spl Split rail check
tar Target R checklist
top Top-down target method
big Trap: big charger, fast charge
bot Trap: both legs in one row
kn Trap: known R in other branch
tr di Trap: LED on a divider
one Trap: one fault only
se le Trap: one series LED reversed
op ba Trap: open-circuit battery V
tro Troubleshooting steps
rou Uncertainty rounding
usb USB label reading
us le USB standard levels
tas USB task checklist

Switches, relays, fuses

trav 3-way common and travelers
wa 3-way switch
loc 4-way switch
ins 4-way switch inside
af AFCI vs GFCI
au Automotive relay pins
bef Break-before-make use
cap Breaking capacity
bre Circuit breaker
con Contact bounce
bl Does the fuse blow
dpd DPDT switch
dp DPST switch
fas Fast vs slow-blow fuse
fl Flyback diode
fu Fuse
2t Fuse heat and I^2t
ty Fuse types
fu vs Fuse vs breaker
forms GFCI forms and history
works GFCI how it works
line GFCI LINE and LOAD
te GFCI TEST button
tri GFCI trip level
gf ti GFCI trip time rule
gf wh GFCI where and limits
lat Latching switch types
mic Limit and micro switch
mu pa Multiway = parity (XOR)
fri NC push button (fridge)
doo NO push button (doorbell)
overl Overload vs short circuit
up Pick-up and drop-out
ptc PTC polyfuse
nc Push button NO vs NC
rc RCD 30 mA
ree Reed switch
re Relay
com Relay COM NO NC states
rati Relay contact rating
compu Relay logic and computers
rese Resetting a breaker
sin Single and double pole
sol Solid-state relay
spd SPDT switch
sp SPST switch
ki Switch kinds
thr Switch poles and throws
ter Switch terminal count
br su Trap: breakers and surges
bac Trap: flyback diode backwards
gf cu Trap: GFCI limits current
pres Trap: NC DPST pressed
overs Trap: oversized fuse/breaker
tr sp Trap: SPST for 2 places
neu Trap: switch on neutral
goe Why a fuse goes in series
prefe Why prefer a fuse
why Why use a relay

Safety

bo cu Body current table
bod Body resistance
go Dalziel AC let-go numbers
dal Dalziel DC numbers
ex Extra-low voltage 50 V
arci Fuse voltage rating: arcing
bon Grounding vs bonding
int bo Internal body resistance
va MOV varistor
pron Polarized and 3-prong plugs
sa Safety habits
fac Shock factors
ai Shock first aid
keye Shock test values keyed
st Step voltage
sur Surge protector
su ra Surge protector ratings
12 Trap: 12 V car battery shock
tet Trap: AC tetanus or DC tetanus
br pr Trap: breakers protect people
hel Why the ground wire helps

Diodes

1n 1N4001 1N4007 1N4148 parts
ba Band gap
ga wa Band gap to wavelength
ge Band gaps: Ge to GaN
br di Bridge diode faults
br rec Bridge rectifier details
dep Depletion region
cla Diode clamp to a reference
curv Diode I-V curve
gates Diode logic gates
mo Diode models
di Diode uses
acc Donors and acceptors
dop Dopant element list
el di Electron flow in a diode
fin Find the cathode
fo Forward bias
four Four regions of diode curve
mino Majority and minority carriers
type N-type and p-type
ph Photodiode facts
piv Rectifier PIV and peaks
re vs Rectifier vs inverter
rec Rectifiers
re bi Reverse bias and breakdown
scho Schottky, Ge, photodiode
sem Semiconductor basics
no Why no silicon LEDs
bea Why silicon beats germanium
ze Zener diode
1n47 Zener parts 1N4733A 1N4742A
av Zener vs avalanche breakdown

LEDs

ant Anti-parallel LED pair
blu Blue and violet LED Vf
gre Green LED: two kinds
makes How an LED makes light
ran LED current range
le LED legs
mate LED materials by color
lit LED not lit trick
le re LED resistor
typ LED typ vs max Vf gap
vf LED Vf by color
cf LED vs CFL vs incandescent
bands LED wavelength bands
ye Orange and yellow LED Vf
63 Red LED Vf 1.63-2.03 V
dim Shunt resistor dims LED
tw Two equal LEDs task
wav Wavelength and Vf
15 White LED and 4.15 V key

Logic

16 16-pin chip power pins
ga ch 3- and 4-input gate chips
7400 7400 quad NAND chip
740 7402 quad NOR chip
7404 7404 hex inverter chip
7408 7408 quad AND chip
741 74138 active-low decoder
742 74266 quad XNOR chip
7428 74283 4-bit adder chip
74 7432 quad OR chip
748 7486 quad XOR chip
abs Absorption
coun Adder gate counts
or na AND and OR from NAND
or no AND and OR from NOR
an AND gate
ans ANSI gate shapes
exa At least k / exactly k
alg Boolean algebra laws
bo id Boolean identities
bu Bubble pushing
pins Chip pins and wiring
ch ro Chips needed: round up
cm CMOS inverter
cm lo CMOS logic levels
comb Combining law
th Consensus theorem
laws De Morgan's laws
dem Demux, decoder, encoder
di ou Diode gate output voltages
do Don't care rows
dua Dual vs complement
en pr Encoder and priority encoder
engl English words to gates
ful Full adder
mi Full adder minterms
ad Half adder
kar Karnaugh map (K-map)
syn Law name synonyms
lo ch Logic chips 7400 series
han Logic hands-on build
lo Logic levels
nota Logic notation symbols
word Logic word problems
maj Majority function
pi Minterm, maxterm, sigma, pi
mu Multiplexer
dec Mux and decoder chips
mu ga Mux as a gate
tr Mux tree N - 1
bigg NAND counts: bigger circuits
na NAND gate
na onl NAND-only gate counts
no na NOR from 4 NANDs
nor NOR gate
on NOR-only gate counts
not NOT gate
or OR gate
or not OR gate from AND and NOT
ord Order of operations

bit Parity bit facts
pri Priority lights
pul Pull-up vs pull-down
ri Ripple-carry adder
sel Select lines: round up
she Sheffer stroke, Peirce arrow
sink Sourcing vs sinking LED
sum Sum of products
tra Trap: + means OR
tr nor Trap: 3-input NOR build
tr xn Trap: 3-input XNOR
tr 74 Trap: 7402 wired like 7400
bo bi Trap: Boolean 1+1 vs binary
buf Trap: buffer does not flip
tr de Trap: De Morgan vs Boole
ha wr Trap: half adder wrong gates
no ti Trap: NAND(A,0) is not NOT
ass Trap: NAND/NOR not assoc.
tr ri Trap: ripple of half adders
tr sh Trap: Shannon vs Boole
sim Trap: simplify before counting
stai Trap: staircase XOR or XNOR
ta or Trap: truth table row order
ta Truth table rows
tt TTL input levels
mar TTL output levels, margins
univ Universal gate sets
unu Unused gate inputs
nors XNOR from 4 NORs
xn XNOR gate
xo bu XOR bubble rules
nands XOR from 4 NANDs
xo XOR gate
id XOR identities

Binary

bin Binary addition
pl Binary place values
nee Bits needed for N
oc Hex, octal, BCD
tr tw Trap: two's complement OUT

Traps

tr si Trap: 1.1 V is not Si Vf
br se Trap: 100 W brighter in series
tr ba Trap: 5-band multiplier
part Trap: ammeter across part
at Trap: ATX 2026 ground key
ol Trap: blue LED reads OL
unk Trap: bridge unknown arm
burn Trap: burns out = open
tr ca Trap: capacitor vs cell
cho Trap: choose the source R
sli Trap: Coulomb unit slips
acr Trap: current "across"
cu us Trap: current is used up
sid Trap: depletion ion sides
di li Trap: diode is non-linear
tr ano Trap: diode vs battery anode
tr ins Trap: E = 0 inside so V = 0
ow Trap: each battery own R
closi Trap: electroscope closing
tr fi Trap: field units N*C
tr bia Trap: forward bias is not on
fu co Trap: fuse count rounds down
gaa Trap: GaAs alone is infrared
gai Trap: gained protons
tr gap Trap: GaP/GaN LED colors
posi Trap: gold by position
tr go Trap: gold is the best
ha vo Trap: half voltage, half power
he se Trap: heat and semiconductor R
tr ci Trap: ideal vs real circuit
tr kc Trap: KCL conserves energy
rev Trap: LED reverse 5 V
look Trap: look-alike symbols
mah Trap: mAh is not energy
po cu Trap: max power vs current
vf li Trap: max Vf is not limit
me pow Trap: meter as power source
minu Trap: minus sign force
nit Trap: N is not nitrogen
neg Trap: negative current answer
watts Trap: no watts in code
tr oh Trap: ohm/m vs ohm m
liv Trap: ohmmeter on live circuit
mode Trap: ohms mode on a diode
of Trap: open means off
2r Trap: P = I^2R for any part
pa bu Trap: parallel bulb removed
varia Trap: pot vs variable R
ov Trap: product over sum
puls Trap: pulsing DC is DC
se sou Trap: R in series w/ I source
ra se Trap: rated watts in series
rod Trap: rod removed first
sam Trap: same reading, 2 faults
tr sc Trap: Schottky is not PN
wa sou Trap: sine wave = AC source
sm Trap: SMD 100 = 10 ohm
tr sup Trap: superposition power
thi Trap: thick wire has more n
sq ch Trap: V = 0 is not E = 0
ac op Trap: V across an open
scala Trap: V is 1/r not 1/r^2
vs th Trap: varistor vs thermistor
wro Trap: yellow Vf key wrong
wired Trap: Zener wired forward

Search codes: index by code

10 pe Trap: is 100 V peak or RMS?
110 US 110, 115, 117, 120 V
12 Trap: 12 V car battery shock
120x Trap: 60 Hz reverses 120x/s
125 Trap: rated 125 V not 120 V
15 White LED and 4.15 V key
16 16-pin chip power pins
17 Volta: methane 1776
178 Coulomb's 1785 data
179 Galvani: 1791 book title
18 Daniell cell 1836
180 Volta letter: 1800 details
182 Ohm's 1826 formula
1827 Ohm: the 1827 controversy
1847 Boole: 1847 book details
185 Timeline: 1850 to 2014
1857 Kirchhoff 1857 signal speed
187 Hertz: 1879 Berlin prize
188 Volt unit named 1881
1887 Hertz: photoelectric 1887
19 19 SI units named for people
194 Shannon: info theory 1948
195 Solar history 1839, 1954
1n 1N4001 1N4007 1N4148 parts
1n47 Zener parts 1N4733A 1N4742A
24 Trap: diesel battery 24 V
240 120 V and 240 V split phase
2r Trap: P = I^2R for any part
2t Fuse heat and I^2t
33 BullSO C 133.33 ohm task
34 Trap: 340 V peak-to-peak
39 Electric vs gravity 10^39
3rd Newton 3rd law: equal force
3s Pack names like 3S2P
60 Japan mains 100 V 50/60 Hz
61 Infinite ladder 1.618R
63 Red LED Vf 1.63-2.03 V
6t Tempco band (6th band)
74 7432 quad OR chip
740 7402 quad NOR chip
7400 7400 quad NAND chip
7404 7404 hex inverter chip
7408 7408 quad AND chip
741 74138 active-low decoder
742 74266 quad XNOR chip
7428 74283 4-bit adder chip
748 7486 quad XOR chip
abs Absorption
ac AC basics
ac op Trap: V across an open
ac or Source: AC or DC?
ac sy AC and DC symbols
acc Donors and acceptors
accu 3.5-digit meter, accuracy
aco Ohm's acoustic law
acr Trap: current "across"
ad Half adder
ad la Adapter label symbols
adds Cells: r in series, parallel
ado Watt and joule adopted 1889
af AFCI vs GFCI
af th After the pile: first uses
aft Units named after people
ai Shock first aid
al Alkaline cell
alg Boolean algebra laws
ali Kirchhoff: law aliases
all Trap: all AC is a sine wave
am Ammeter
am ba Ampere is the base unit
am le Trap: ammeter 0, LED lit
am na Ampere named galvanometer
am si Ampere: only SI base unit
am un Ampere unit history
ampe Ampere: life
an AND gate
ano Anode and cathode
ans ANSI gate shapes
ant Anti-parallel LED pair
ap Typical appliance powers
ar DC arcs harder to stop
arci Fuse voltage rating: arcing
as Ampere: astatic needle
ass Trap: NAND/NOR not assoc.
at Trap: ATX 2026 ground key
att Coulomb: attraction method
au Automotive relay pins
av Zener vs avalanche breakdown
ba Band gap
ba co Battery symbol: count pairs
ba ho Torsion balance: how it works
bac Trap: flyback diode backwards
bala Balanced Wheatstone bridge
bands LED wavelength bands
bank Power bank real output
bc Timeline: 600 BC to 1850
be Pith ball behavior
bea Why silicon beats germanium
bef Break-before-make use
ben Bench supply CV and CC
big Trap: big charger, fast charge
bigg NAND counts: bigger circuits
bin Binary addition
bis Trap: Ohm's copper-bismuth
bit Parity bit facts
bl Does the fuse blow
bl no Blue LED Nobel 2014 detail
bla Kirchhoff: black body
blu Blue and violet LED Vf
bo Boole: life
bo bi Trap: Boolean 1+1 vs binary
bo cu Body current table
bo id Boolean identities
bo la Boole's law x^2 = x
bod Body resistance
bon Grounding vs bonding
books Boole: books
bot Trap: both legs in one row
box Black box: find the diode
br di Bridge diode faults
br pr Trap: breakers protect people
br rec Bridge rectifier details
br ro Breadboard rows of 5
br se Trap: 100 W brighter in series
br su Trap: breakers and surges
bra Shorted parallel branch
bre Circuit breaker
bu Bubble pushing
bu ra Bulb rating to resistance
buc Buck, boost, DC-DC
buf Trap: buffer does not flip
bul Bulb brightness
bun Stretch or bundle a wire
bur Ammeter burden
burn Trap: burns out = open
ca en Capacity and energy
cag Faraday cage: charge inside
cap Breaking capacity
car Car safe in lightning
cau Static shock cause
cav Cavendish found it first
cc CC/CV slow finish
cells Cells in series and parallel
ces Kirchhoff: cesium and rubidium
cf LED vs CFL vs incandescent
ch Charge time
ch ro Chips needed: round up
cho Trap: choose the source R
cit Kirchhoff: cities
cla Diode clamp to a reference
clo Pick R_k close to R_x
closi Trap: electroscope closing
cm CMOS inverter
cm lo CMOS logic levels
co Commutator
co br Coin bridge switch test
co fl Conventional vs electron flow
co hi Coulomb's law history
co la Coulomb's law
co pr Before Coulomb: priority
co un Coulomb unit = 1 A*s
coe Temperature coefficient
com Relay COM NO NC states
comb Combining law
commo Find the SPDT common pin
compu Relay logic and computers
con Contact bounce
cond Conductors in fields
cong Units at 1881 Paris congress
cons Charge constants
conti Continuity and diode test
cop AWG copper table
cos Electricity cost
cou Coulomb: life
coun Adder gate counts
cr CR2032 lithium coin
cu Current
cu di Current divider
cu us Trap: current is used up
cub Resistor cube values
curv Diode I-V curve
cut Symmetry: join or cut
da AC vs DC danger
dal Dalziel DC numbers
dc DC basics
dc us DC sources and uses
de De Morgan: life
de le Boole and De Morgan letters
dea Dead lamp: 0 V or full V
dec Mux and decoder chips
del Delta-Y conversion
deli Absorb or deliver power
dem Demux, decoder, encoder
dep Depletion region
det Shannon thesis details
di Diode uses
di li Trap: diode is non-linear
di ou Diode gate output voltages
di te Diode test readings
die Ohm: Die galvanische Kette
dig Color code digits
dim Shunt resistor dims LED
dir Force direction in a field
dis Ampere's discoveries
disch Discharge types order
dist Coulomb ratio: distance
div Voltage divider
do Don't care rows
doo NO push button (doorbell)
dop Dopant element list
dp DPST switch
dpd DPDT switch
dr Dry cell vs wet cell
dri Drift speed of electrons
du Du Fay: two kinds of charge
dua Dual vs complement
e1 E12 standard values
e6 E6 and E24 series
ea Earth ground symbol
ed Edison's argument vs AC
ef Efficiency
el Electron-volt
el ch Electric charge basics
el di Electron flow in a diode
el fi Electric field
el ho Electrophorus: how it works
el ind Electroscope by induction
el ke Trap: electrodynamics key
em EMF is not a force
en Energy and kWh
en co Energy conversions
en pr Encoder and priority encoder
end End of the war 1892
engl English words to gates
eq Equivalent resistance tips
eq va Equal-resistor values
equi Equipotential rules
es ESD you cannot feel
eve Everyday source voltages
ex Extra-low voltage 50 V
exa At least k / exactly k
exp Volta quote: experiment
fa Fault clues
fac Shock factors
far Finite ladder: far end first
fara Trap: Oersted law = Faraday
fas Fast vs slow-blow fuse
fe Resistivity: W, Fe, Pt, Pb
fi Field lines
fig Sig fig counting rules
fil Bulb filament and inrush
fin Find the cathode
fixe Battery to ground fixes a node
fl Flyback diode
fla Flash-to-bang rule
flo Grounding electron flow
flu Trap: Ampere and fluorine
fo Forward bias
fo ma Trap: Ampere found magne…
forms GFCI forms and history
fou Ohm: inspired by Fourier
four Four regions of diode curve
fr Franklin: battery and Fr unit
fra Franklin
fri NC push button (fridge)
fu Fuse
fu co Trap: fuse count rounds down
fu sy Fuse and breaker symbols
fu vs Fuse vs breaker
ful Full adder
ga Galvani
ga ch 3- and 4-input gate chips
ga wa Band gap to wavelength
gaa Trap: GaAs alone is infrared
gai Trap: gained protons
galvano Galvanometer
gates Diode logic gates
ge Band gaps: Ge to GaN
gf cu Trap: GFCI limits current
gf ti GFCI trip time rule
gf wh GFCI where and limits
gi Gilbert: De Magnete 1600
gl Crown of cups: wine glasses
go Dalziel AC let-go numbers
goe Why a fuse goes in series
gol Color code gold silver
gr Slope of V-I graph
grad Coulomb quote: graduating
gre Green LED: two kinds
gri Why the grid uses AC
grov Fuel cell: Grove 1839
gu Gray and von Guericke
ha DC hazards
ha vo Trap: half voltage, half power
ha wr Trap: half adder wrong gates
han Logic hands-on build
he Ampere: words he coined
he se Trap: heat and semiconduct…
he tr Hertz: the transmitter
hel Why the ground wire helps
her Hertz: life
hi Ohm's law history
ho Ohm: honours
hol Holonyak: first visible LED

`honor` Volta honors timeline
`hot` Hot, neutral, ground roles
`hou` Household wire colors
`hp` Watt: steam engine, hp
`hu` Humidity and static
`hug` Huge mystery R: loading
`hv` HVDC lines
`id` XOR identities
`idea` Ideal meters memory trick
`ie` ANSI vs IEC symbols
`ig` Fuel ignition risk
`ind` Indirect measurements
`indu` Induction steps in order
`ins` 4-way switch inside
`insu` Conductors vs insulators
`int` Internal resistance
`int bo` Internal body resistance
`inven` Trap: who invented AC
`io` Lithium-ion cell
`ite` Infinite ladder 0.618R
`ja` Multimeter jacks
`jack` Fuse in the mA jack
`je` Boole's OR vs Jevons
`jo` Joule and Watt
`jo oh` Trap: Joule before Ohm
`joi` Joined spheres: Q ~ r
`ju` Franklin: kite June 1752
`kar` Karnaugh map (K-map)
`kc` KCL current law
`ke` Kelvin bridge
`keye` Shock test values keyed
`ki` Switch kinds
`ki di` Kirchhoff's discoveries
`ki th` Kirchhoff: the other laws
`kir` Kirchhoff: life
`kn` Trap: known R in other branch
`kv` Air breakdown 3 kV/mm
`kv la` KVL voltage law
`la` Ohm's law
`lam` Lamp symbol: X or loop
`lat` Latching switch types
`laws` De Morgan's laws
`lay` Breadboard layout
`ld` Thermistor, varistor, LDR
`le` LED legs
`le ch` Equal LEDs checklist
`le fa` LED fault from voltages
`le re` LED resistor
`le ze` Lead resistance zero
`lea` Lead-acid cell
`leaf` Electroscope leaf rules
`lef` Trap: lead left in mA jack
`lem` Lemon or potato cell
`len` Ampere force law: lengths
`les` Trap: AC has less resistance
`lett` Meter symbols: letters
`ley` Pile vs Leyden jar
`li` Ohm: life
`lim` Breadboard limits
`line` GFCI LINE and LOAD
`lip` LiPo and LiFePO4
`lit` LED not lit trick
`liv` Trap: ohmmeter on live circuit
`lo` Logic levels
`lo ch` Logic chips 7400 series
`loa` Loading effect
`loade` Loaded divider: output drops
`loc` 4-way switch
`look` Trap: look-alike symbols
`los` Line loss and voltage
`mad` Trap: Volta made electrophorus
`mag` Trap: On Magnetism key
`mah` Trap: mAh is not energy
`mai` Mains in other countries
`maj` Majority function
`makes` How an LED makes light
`mar` TTL output levels, margins
`mate` LED materials by color
`max` Max power transfer
`maxw` Maxwell: link to Hertz
`me` Mesh analysis
`me po` Meter polarity: minus sign
`me pow` Trap: meter as power source
`mem` Ohm quote: the Memoir
`meta` Resistivity: non-metals
`metal` Metal resistivity order
`meth` Charging methods
`mh` Mho and siemens history
`mi` Full adder minterms
`mic` Limit and micro switch
`mil` One-node formula (Millman)
`mini` Minimum safe resistor
`mino` Majority and minority carriers
`minu` Trap: minus sign force
`mix` Trap: Galvani mix-ups
`mn` Color code mnemonic
`mo` Diode models
`mode` Trap: ohms mode on a diode
`moh` Wrist strap 1 Mohm resistor
`mos` Tribo: most charge pair
`mu` Multiplexer
`mu ga` Mux as a gate
`mu pa` Multiway = parity (XOR)
`my` Mystery R checklist
`my re` Mystery resistor task
`na` NAND gate
`na onl` NAND-only gate counts
`nai` Trap: two copper nails = 0 V
`names` Seven names mnemonic
`nands` XOR from 4 NANDs
`nc` Push button NO vs NC
`ne` NEMA name code
`nee` Bits needed for N
`neg` Trap: negative current answer
`neu` Trap: switch on neutral
`nev` Trap: wave never below zero
`new` Newton of electricity quote
`ni` NiCd cell
`nich` Why heaters use nichrome
`nim` NiMH cell
`nit` Trap: N is not nitrogen
`no` Why no silicon LEDs
`no na` NOR from 4 NANDs
`no ti` Trap: NAND(A,0) is not NOT
`nob` Trap: Nobel starts 1901
`nod` Nodal analysis
`nodes` Series vs parallel by nodes
`non` Non-SI conversions
`nor` NOR gate
`nors` XNOR from 4 NORs
`not` NOT gate
`nota` Logic notation symbols
`nu` Lightning numbers
`oc` Hex, octal, BCD
`oe` Oersted
`of` Trap: open means off
`oh` Ohmmeter
`oh un` Ohm unit name history
`ohmi` Ohmic vs non-ohmic
`ol` Trap: blue LED reads OL
`on` NOR-only gate counts
`one` Trap: one fault only
`ont` Schematic onto breadboard
`op` Open circuit
`op ba` Trap: open-circuit battery V
`opens` Switch closes or opens
`or` OR gate
`or na` AND and OR from NAND
`or no` AND and OR from NOR
`or not` OR gate from AND and NOT
`ord` Order of operations
`ot` Volta: other work
`ou` Outlet slots
`ou me` Outlet meter readings
`ov` Trap: product over sum
`overl` Overload vs short circuit
`overs` Trap: oversized fuse/breaker
`ow` Trap: each battery own R
`pa` How parts fail
`pa bu` Trap: parallel bulb removed
`pa ce` Parts of a cell
`pa dimm` Bulb A then B, C in parallel
`pa ra` Series / parallel ratio n^2
`pad` Joule: paddle wheel
`pai` Prefix shortcut pairs
`pap` Oersted: paper and Arago
`para` Parallel resistors
`part` Trap: ammeter across part
`pc` PCB printed circuit board
`pe` Zener: person vs diode
`ph` Photodiode facts
`pi` Minterm, maxterm, sigma, pi
`pic` Pick the best range
`pil` Voltaic pile
`pins` Chip pins and wiring
`piv` Rectifier PIV and peaks
`pl` Binary place values
`plu` Trap: S vs s, plurals
`pn` Ohl: PN junction details
`pn le` PN junction and LED history
`po` Polarization
`po ch` Power task checklist
`po cu` Trap: max power vs current
`po en` Potential and energy
`po rh` Potentiometer vs rheostat
`pog` Ohm: Poggendorff thermocou…
`poi` Hands-on scoring points
`posi` Trap: gold by position
`post` Ampere: career posts
`posth` Coulomb unit: posthumous
`pot` Potentiometer
`pow` Power formulas
`pr ce` Primary cell voltages
`pred` Trap: predicted vs confirmed
`prefe` Why prefer a fuse
`pres` Trap: NC DPST pressed
`pri` Priority lights
`prim` Primary vs secondary
`pron` Polarized and 3-prong plugs
`prop` Volta's law: Q prop. to V
`prov` Coulomb: proved vs assumed
`ptc` PTC polyfuse
`pul` Pull-up vs pull-down
`puls` Trap: pulsing DC is DC
`qs` Volta vs Galvani: past Qs
`qu` Hertz quotes
`qu un` Quantities and units table
`qua` Quark charges
`quan` Charge quantization
`ra` C-rate
`ra se` Trap: rated watts in series
`rad` Hertz: radio waves
`ran` LED current range
`rank` Identical bulbs: rank rules
`rati` Relay contact rating
`ratio` Coulomb ratio: charges
`ray` Hertz: cathode rays
`rc` RCD 30 mA
`re` Relay
`re ba` Reading 4, 5, 6 bands
`re bi` Reverse bias and breakdown
`re ch` Resistor tolerance check
`re po` Resistor power rating
`re va` Resistivity values
`re vs` Rectifier vs inverter
`read` Which end to read first
`rec` Rectifiers
`rece` Hertz: the receiver
`rech` Rechargeable cell voltages
`red` Red lead is anode
`ree` Reed switch
`res` Temperature and resistance
`rese` Resetting a breaker
`resi` Resistivity
`rev` Trap: LED reverse 5 V
`rh` Rheostat
`ri` Ripple-carry adder
`riv` Trap: Ohm's rival not Galvani
`rm` RMS
`rod` Trap: rod removed first
`rou` Uncertainty rounding
`roy` Boole: Royal Medal 1844
`ru` KVL sign rules
`sa` Safety habits
`sam` Trap: same reading, 2 faults
`sc` Out of scope 2027
`sc ba` Schematic symbols: basics
`sca` AWG scaling rules
`scala` Trap: V is 1/r not 1/r^2
`sch` Schematic symbols: parts
`scho` Schottky, Ge, photodiode
`sci` The seven 2027 scientists
`se` Meter settings
`se le` Trap: one series LED reversed
`se sou` Trap: R in series w/ I source
`seeb` Thermocouple: Seebeck 1821
`sel` Select lines: round up
`sem` Semiconductor basics
`sep` Ampere: Sept 1820 timeline
`ser` Series resistors
`sev` Seven SI base units
`sh` Short circuit
`sh se` Shorting one series bulb
`shan` De Morgan and Shannon
`shar` Hands-on share of score
`she` Sheffer stroke, Peirce arrow
`show` Hertz: wave properties shown
`shu` Shunt vs multiplier
`si` SI prefixes
`si oh` SI base units of V, ohm, S
`sid` Trap: depletion ion sides
`sig` Sig figs in math
`sim` Trap: simplify before counting
`sin` Single and double pole
`sink` Sourcing vs sinking LED
`sli` Trap: Coulomb unit slips
`sm` Trap: SMD 100 = 10 ohm
`so` Sources of error
`soc` Lamp socket: shell is neutral
`sod` Trap: sodium in Sun, not Moon
`sol` Solid-state relay
`sola` Solar cell numbers
`sp` SPST switch
`spd` SPDT switch
`spect` Kirchhoff: spectroscope trap
`spl` Split rail check
`sq` Trap: RMS of a square wave
`sq ch` Trap: V = 0 is not E = 0
`st` Step voltage
`sta` Static hazards
`stai` Trap: staircase XOR or XNOR
`stead` Steady-state laws multi-sel…
`steps` Superposition steps
`stor` Trap: battery stores charge
`str` Power limit of a string
`stru` Trap: kite struck by lightning
`stu` Kirchhoff: 1845 as a student
`su` Supernode
`su di` Superconductor discovery
`su ra` Surge protector ratings
`sum` Sum of products
`sun` Kirchhoff quote: gold sun
`supp` Power supplied task
`sur` Surge protector
`swa` Edison: bulb 1879 and Swan
`sy` Boole's symbols
`sy li` Anti-symmetry: V/2 line
`syn` Law name synonyms
`ta` Truth table rows
`ta or` Trap: truth table row order
`tap` Pot taper: linear vs log
`tar` Target R checklist
`tas` USB task checklist
`te` GFCI TEST button
`ter` Switch terminal count
`tesl` Tesla (out for 2027)

tet Trap: AC tetanus or DC tetanus
th Consensus theorem
th sy Thermistor and LDR symbols
tha Thales and amber c. 600 BC
thev Thevenin and Norton: OUT
thi Trap: thick wire has more n
tho Boole: Laws of Thought title
thr Switch poles and throws
thu Thunderstorm charging
timel Timeline of the seven
to Network tolerance
top Top-down target method
tor Trap: torsion balance vs Ohm
tou Identical spheres touch
tr Mux tree N - 1
tr 74 Trap: 7402 wired like 7400
tr ad Trap: adapter vs converter
tr am Trap: Coulomb or Ampere?
tr ano Trap: diode vs battery anode
tr ba Trap: 5-band multiplier
tr bia Trap: forward bias is not on
tr ca Trap: capacitor vs cell
tr ci Trap: ideal vs real circuit
tr de Trap: De Morgan vs Boole
tr di Trap: LED on a divider
tr el Trap: 1 A = 1 electron/s
tr fi Trap: field units N*C
tr gap Trap: GaP/GaN LED colors
tr go Trap: gold is the best
tr ins Trap: E = 0 inside so V = 0
tr ju Trap: junction rule = charge
tr kc Trap: KCL conserves energy
tr ki Trap: not Kirchhoff's work
tr la Trap: Coulomb law vs unit
tr nor Trap: 3-input NOR build
tr oh Trap: ohm/m vs ohm m
tr pa Trap: 1/8 W parts
tr ri Trap: ripple of half adders
tr sc Trap: Schottky is not PN
tr sh Trap: Shannon vs Boole
tr si Trap: 1.1 V is not Si Vf
tr sp Trap: SPST for 2 places
tr sup Trap: superposition power
tr tw Trap: two's complement OUT
tr vo Trap: not Volta's
tr wa Trap: watt vs joule
tr wo Trap: not Boole's work
tr xn Trap: 3-input XNOR
tra Trap: + means OR
trav 3-way common and travelers
trea Boole quote: treatise
tri GFCI trip level
trib Triboelectric series
tro Troubleshooting steps
tt TTL input levels
tw Two equal LEDs task
tw sp Two touching spheres indu…
ty Fuse types
ty ba Typical battery capacities
typ LED typ vs max Vf gap
type N-type and p-type
typi Typical currents
un k value and units
un eq Units equal to a volt
un pi Count unknowns, pick met…
un wa Units equal to a watt or ohm
unb Unbalanced bridge voltmeter
unc Carry uncertainty
unif Uniform field E = V/d
unit Hertz unit history
univ Universal gate sets
unk Trap: bridge unknown arm
unu Unused gate inputs
up Pick-up and drop-out
us US and world outlets
us le USB standard levels
usb USB label reading
used Trap: copper is used up
va MOV varistor
van Van de Graaff safe to touch
varia Trap: pot vs variable R
varie Rail polarity varies
ve Tribo series: long version
verso Gilbert: electricus, versorium
vf LED Vf by color
vf li Trap: max Vf is not limit
vo Voltage
vo bu Voltaic pile: build details
vo co Volta's contact series
vo se Voltmeter in series
vo vs Volta vs Galvani: details
volta Volta: life
voltm Voltmeter
vs th Trap: varistor vs thermistor
wa 3-way switch
wa bu 3-way build checklist
wa fa 3-way fault tables
wa sou Trap: sine wave = AC source
wal Wall adapter insides
war War of the Currents
watts Trap: no watts in code
wav Wavelength and Vf
wea Weak battery reads full
wer Werner von Siemens
wes Westinghouse: AC wins
wh el Trap: who coined electricity
wha Shorted bulb: what happens
whe Field where E = 0
whea Wheatstone bridge history
whi Trap: which Siemens
whil Joule: unit named while alive
why Why use a relay
wi Wire gauge AWG
wim Wimshurst machine
wired Trap: Zener wired forward
word Logic word problems
worke Trap: Volta worked in France
works GFCI how it works
wr Unit writing rules
wro Trap: yellow Vf key wrong
xn XNOR gate
xo XOR gate
xo bu XOR bubble rules
ye Orange and yellow LED Vf
ze Zener diode
ze cu Zero-current branch trick
ze po Zero-potential points: two
zer 3-band and zero-ohm
zi Carbon-zinc (Leclanche)